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Liveness detection method and system thereof

Abstract

A liveness detection method includes: obtaining multiple first feature points and multiple second feature points according to a first image, a second image, and a feature point process, wherein the first feature points include two first fixed feature points and at least one first variable feature point, and the second feature points include two second fixed feature points and at least one second variable point; obtaining a first transform function according to the first fixed feature points and the second fixed feature points; obtaining at least one check feature point according to the first transform function and the first variable feature point(s); and determining whether the second image is a spoofing image according to the check feature point(s) and the second variable point(s).

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This non-provisional application claims priority under 35 U.S.C. § 119(a) to Patent Application No. 110148551 filed in Taiwan, R.O.C. on Dec. 23, 2021, the entire contents of which are hereby incorporated by reference.

BACKGROUND

Technical Field

(2) The present application relates to a facial recognition technology, in particular, a liveness detection method and system thereof.

Related Art

(3) Various applications of facial recognition technology are gradually increasing, and considering that different usage scenarios have different requirements for privacy and security levels, the application requirements of anti-spoofing technology, such as liveness detection, are born

accordingly. In this way, not only biological features (such as facial features) can be detected, but also whether a person in the image to be recognized is a true person or not can be identified, thereby the occurrence of using photos or videos for spoofing can be avoided.

(4) However, an edge device equipped with a facial recognition function generally has fewer computing resources and memory resources, and the facial recognition function has occupied most of the computing resources and memory resources of the edge device. Therefore, if a liveness detection function requires greater computing resources and memory resources, it may cause the edge device to be unable to support the computing resources and memory resources required by the liveness detection function. In other words, the edge device may not be able to load the liveness detection function, thereby the occurrence of spoofing cannot be prevented.

SUMMARY

(5) In view of the above, a liveness detection method and system thereof are provided in the present application. According to some embodiments, the liveness detection method includes: obtaining two first fixed feature points and at least one first variable feature point according to a first image and a feature point process; obtaining two second fixed feature points and at least one second variable feature point according to a second image and the feature point process; obtaining a first transform function according to the first fixed feature points and the second fixed feature points; obtaining at least one check feature point according to the first transform function and the at least one variable feature points; and determining whether the second image is a spoofing image according to the at least one check feature point and the at least one second variable feature point.

(6) According to some embodiments, a liveness detection system includes a non-transitory storage medium and a processor. The processor is electrically connected to the non-transitory storage medium. The non-transitory storage medium is configured to store a program. The processor is configured to load the program to perform the following steps: obtaining two first fixed feature points and at least one first variable feature point according to a first image and a feature point process; obtaining two second fixed feature points and at least one second variable feature point according to a second image and the feature point process; obtaining a first transform function according to the first fixed feature points and the second fixed feature points; obtaining at least one check feature point according to the first transform function and the at least one first variable feature point; and determining whether the second image is a spoofing image according to the at least one check feature point and the at least one second variable feature point.

(7) According to some embodiments, a liveness detection method includes: obtaining a plurality of first feature points according to a first image and a feature point process; obtaining a plurality of second feature points according to a second image and the feature point process; obtaining a fourth transform function according to the first feature points and the second feature points; and determining the second image is a liveness image when the fourth transform function does not meet a predetermined constraint, wherein the predetermined constraint comprises an epipolar constraint and an affine constraint.

(8) To sum up, according to some embodiments, the first transform function and the fourth transform function can be obtained through a simple operation, and whether a person image to be recognized is a spoofing image can be determined according to the first transform function and the fourth transform function, thereby determining whether the behavior of using photos or videos for spoofing occurs. In other words, while realizing the determination of the spoofing behavior, the computing resources and memory resources required for performing the determination of the spoofing behavior can also be reduced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The disclosure will become more fully understood from the detailed description given herein below for illustration only, and thus not limitative of the disclosure, wherein:
- (2) FIG. 1 illustrates a schematic block diagram of a liveness detection system according to some embodiments of the present application;
- (3) FIG. 2 illustrates a schematic flow diagram of a liveness detection method according to some embodiments of the present application;
- (4) FIGS. 3A-3B illustrate a schematic diagram of a first original image and a second original image according to some embodiments of the present application;
- (5) FIG. 4 illustrates a schematic diagram of the conversion of a first transform function according to some embodiments of the present application;
- (6) FIG. 5 illustrates a schematic flow diagram of a liveness detection method according to some embodiments of the present application;
- (7) FIG. 6 illustrates a schematic flow diagram of a liveness detection method according to some embodiments of the present application;
- (8) FIG. 7 illustrates a schematic flow diagram of a liveness detection method according to some embodiments of the present application;
- (9) FIG. 8 illustrates a schematic flow diagram of a deskew process according to some embodiments of the present application; and
- (10) FIG. 9A to FIG. 9C illustrates a schematic diagram of an original image and a deskewed image according to some embodiments of the present application.

DETAILED DESCRIPTION

- (11) Refer to FIG. 1, illustrating a schematic block diagram of a liveness detection system **10** according to some embodiments of the present application. The liveness detection system **10** includes a non-transitory storage medium **11** and a processor **13**. In some embodiments, the liveness detection system **10** further includes an image capture device **15**. The processor **13** is electrically connected to the non-transitory storage medium **11** and the image capture device **15**. The image capture device **15** is configured to photograph a user (specifically, the user's face) to generate a video signal of a set of continuous frames. For example, the video signal is 60 continuous frames per second. Wherein, each frame is an original image **40** (as shown in FIG. 9A). The non-transitory storage medium **11** stores a program for the processor **13** to perform, according to the frames, the liveness detection method of the present application after loading. In this way, whether the user is a true person or a fake person is determined. Wherein, the fake person may be a user who uses a mask to fake or a user who uses a photo to fake, and the true person refers to the user himself/herself.
- (12) In some embodiments, the process **13** may be an operation circuit such as a central processing unit, a microprocessor, an application-specific integrated circuit (ASIC), or a system on a chip (SOC).
- (13) Refer to FIG. 2, illustrating a schematic flow diagram of a liveness detection method according to some embodiments of the present application. First, the processor **13** obtains a plurality of first feature points according to a first image (hereinafter referred to as a first original image **20**, as shown in FIG. 3A) and a feature point process (step S201) and obtains a plurality of second feature points according to a second image (hereinafter referred to as a second original image **30**, as shown in FIG. 3B) and the feature point process (step S203). The plurality of first feature points include two first fixed feature points and at least one first variable feature point. The plurality of second feature points include two second fixed feature points and at least one second variable feature point. The first original image **20** may be the first frame among the aforementioned continuous frames, and any one of the remaining successive frames may be the second original image **30**, but the present application is not limited thereto. The first original image **20** may be one

frame whose order after the second one among the aforementioned continuous frames, and any frame following the first original image **20** may be the second original image **30**.

(14) The feature point process can be implemented by a convolutional neural network (CNN). For example, the designer can input multiple sample images including a human face to the processor **13** through an I/O interface (such as a keyboard, a mouse, an image transmission interface, etc.) (not shown). The processor **13** performs, according to the sample images, a machine learning training of the image features (such as histograms of oriented gradients (HOG) features, Haar-like feature (Haar), and so on) to determine determination logic (hereinafter referred to as first determination logic). The machine learning training of the image features related to face detection may be a known or self-developed model, and the details are omitted here. The processor **13** can perform, according to the first determination logic, face detection on the first original image **20** and the second original image **30** to determine whether there is a human face in the first original image **20** and the second original image **30** and outlines the face positions in the first original image **20** and the second original image **30** (hereinafter the first original image **20** with the face position be outlined is referred to as a first outlined image, and the second original image **30** with the face position be outlined is referred to as a second outlined image). The designer can also input multiple sample images with the face position being outlined (hereinafter referred to as face-outlined sample images) to the processor **13** through the I/O interface. The processor **13** performs the machine learning training of the feature points according to the face-outlined sample images to determine determination logic (hereinafter referred to as second determination logic). The machine learning training of the feature points may be a known or self-developed model, and the details are omitted here. The processor **13** can detect, according to the second determination logic, the feature points of the human face in the images (e.g., the first outlined image and the second outlined image). The feature points, such as but are not limited to eyebrow peaks, eyeballs, nostrils, the middle of the upper lip, the middle of the lower lip, etc.

(15) Compared with the first variable feature points and the second variable feature points, the first fixed feature points and the second fixed feature points are less susceptible to change with expressions. For example, the first fixed feature points and the second fixed feature points may be the eyeballs, nostrils, and the middle of the upper lip, and the first variable feature points and the second variable feature points may be the eyebrow peaks and the middle of the lower lip. In some embodiments, the first fixed feature points and the second fixed feature points correspond to the same feature of the face. For example, the two first fixed feature points and the two second fixed feature points both correspond to the two eyeballs of the face. The first variable feature points and the second variable feature points may correspond to the same feature of the face. For example, the first variable feature points and the second variable feature points both correspond to the same eyebrow peaks of the face.

(16) In some embodiments, the sequence of step **S203** and step **S201** may be reversed. In this embodiment, before performing step **S201**, the processor **13** can store the first original image **20** in the non-transitory storage medium **11** for the processor **13** to perform step **S201** subsequently.

(17) In some embodiments, the number of the first variable feature points is the same as or different from the number of the second variable feature points. In some embodiments, the number of the first variable feature points and the number of the second variable feature points are both greater than a number threshold. In some embodiments, if the number of the first variable feature points is not greater than the number threshold, the processor **13** controls the image capture device **15** to generate another video signal, having a new set of continuous frames (hereinafter referred to as another set of frames), and the processor **13** re-performs the feature point process on a first original image **20** in another set of frames until the number of the first variable feature points obtained from the single first original image **20** is greater than the number threshold. In some embodiments of step **S203**, under the condition that the number of the first variable feature points is greater than the number threshold, if the number of the second variable feature points obtained from the second

image **30** is not greater than the number threshold, the processor **13** repeats selecting another second original image **30** from the frames following the first original image **20**, and performs the feature point process on another second original image **30** until the number of the second variable feature points obtained from the single second original image **30** is greater than the number threshold. In a single set of frames, under the condition that all the frames following the first original image **20** have been selected, when the number of the second variable feature points obtained from the single second original image **30** is still not greater than the number threshold, the processor **13** controls the image capture device **15** to generate another video signal having another set of frames, and the processor **13** repeats step **S201** according to another set of frames. The number threshold may be stored in the non-transitory storage medium **11** in advance.

(18) Refer again to FIG. 2. After obtaining the first fixed feature points, the at least one first variable feature point, the second fixed feature points, and the at least one second variable feature point, the processor **13** obtains a transform function (hereinafter referred to as a first transform function) according to the first fixed feature points and the second fixed feature points (step **S205**). For example, the processor **13** can perform translation operation, rotation operation, and scaling operation on the coordinate vectors of the first fixed feature points so as to convert the first fixed feature points to the second fixed feature points, and integrates the translation operation, the rotation operation, and the scaling operation to form the first transform function.

(19) Refer to FIGS. 3A-3B and FIG. 4. FIGS. 3A-3B illustrate a schematic diagram of a first original image **20** and a second original image **30** according to some embodiments of the present application. FIG. 4 illustrates a schematic diagram of the conversion of a first transform function according to some embodiments of the present application. In some embodiments, the first transform function can be represented by equation 1. Equation 1 can be derived from equation 2 to equation 8. Wherein, O.sub.S and A.sub.S are the first fixed feature points from the first original image **20**, O.sub.D and A.sub.D are the second fixed feature points from the second original image **30**, t.sub.x is a horizontal translation amount of O.sub.S and A.sub.S, and t.sub.y is a vertical translation amount of O.sub.S and A.sub.S.  is the vector of  after translation.  is the vector of  after rotation, and ϕ' is the angle between  and .  is then converted to  after being rotated and scaled. ϕ'' is the angle between  and . ds.sub.x is a horizontal component of the vectors of O.sub.S and A.sub.S, ds.sub.y is a vertical component of the vectors of O.sub.S and A.sub.S, dd.sub.x is a horizontal component of the vectors of O.sub.D and A.sub.D, and dd.sub.y is a vertical component of the vectors of O.sub.D and A.sub.D. x is the horizontal axis, and y is the vertical axis.

$$(20) \quad \begin{bmatrix} rs0 & -rs1 \\ rs1 & rs0 \end{bmatrix} \begin{bmatrix} t_x \\ t_y \end{bmatrix} \text{ Equation(1)} = \begin{bmatrix} t_x \\ t_y \end{bmatrix} \text{ Equation(2)}$$

$$= \begin{bmatrix} \cos\phi' & -\sin\phi' \\ \sin\phi' & \cos\phi' \end{bmatrix} \text{ Equation(3)} = \begin{bmatrix} S & 0 \\ 0 & S \end{bmatrix} \begin{bmatrix} \cos\phi'' & \sin\phi'' \\ -\sin\phi'' & \cos\phi'' \end{bmatrix} \text{ Equation(4)}$$

$$\begin{bmatrix} rs0 & -rs1 \\ rs1 & rs0 \end{bmatrix} = \begin{bmatrix} S & 0 \\ 0 & S \end{bmatrix} \begin{bmatrix} \cos\phi'' & \sin\phi'' \\ -\sin\phi'' & \cos\phi'' \end{bmatrix} \begin{bmatrix} \cos\phi' & -\sin\phi' \\ \sin\phi' & \cos\phi' \end{bmatrix} \text{ Equation(5)}$$

$$S = \frac{\sqrt{dd_x^2 + dd_y^2}}{\sqrt{ds_x^2 + ds_y^2}} \text{ Equation(6)} \quad rs0 = \frac{ds_x * dd_x + ds_y * dd_y}{ds_x^2 + ds_y^2} \text{ Equation(7)}$$

$$rs1 = \frac{ds_x * dd_y - ds_y * dd_x}{ds_x^2 + ds_y^2} \text{ Equation(8)}$$

(21) Refer again to FIG. 2. After obtaining the first transform function, the processor **13** obtains at least one check feature point according to the first transform function and the at least one first variable feature point (step **S207**). Specifically, through the first transform function, the first

variable feature points are converted to the check feature points. In this way, the check feature points and the second variable feature points are under the same coordinate reference. In some embodiments, there is a first distance between the two first fixed feature points and a second distance between the two second fixed feature points. The first distance is the same as or different from the second distance. Since the first transform function can be a vector function, the accuracy of the conversion of the first variable feature point to the check feature point by the first transform function can be improved through the first distance and the second distance.

(22) In some embodiments, the first transform function can be represented by equation 9, and equation 1 is substantially the same as equation 9. Wherein, $rs0$, $rs1$, $t_{sub.x}$, and $t_{sub.y}$ are the same as those in equation 1 and will not be repeated. $dc_{sub.x}$ is a horizontal coordinate of the at least one check feature point, $dc_{sub.y}$ is a vertical coordinate of the at least one check feature point, $sc_{sub.x}$ is a horizontal coordinate of the at least one first variable feature point, and $sc_{sub.y}$ is a vertical coordinate of the at least one first variable feature point.

$$(23) \begin{bmatrix} dc_x \\ dc_y \end{bmatrix} = \begin{bmatrix} rs0 & -rs1 \\ rs1 & rs0 \end{bmatrix} \begin{bmatrix} sc_x \\ sc_y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix} \quad \text{Equation(9)}$$

(24) In some embodiments, the first transform function can be represented by equation 10, and equation 10 is substantially the same as equation 9. Wherein, $rs0$, $rs1$, $t_{sub.x}$, $t_{sub.y}$, $dc_{sub.x}$, $dc_{sub.y}$, $sc_{sub.x}$, and $sc_{sub.y}$ are the same as those in equation 9 (or equation 1) and will not be repeated.

$$(25) \begin{bmatrix} dc_x \\ dc_y \\ 1 \end{bmatrix} = \begin{bmatrix} rs0 & -rs1 & t_x \\ rs1 & rs0 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} sc_x \\ sc_y \\ 1 \end{bmatrix} \quad \text{Equation(10)}$$

(26) After obtaining the at least one check feature point, the processor **13** determines whether the second original image is a spoofing image according to the at least one check feature point and at least one second variable feature point (step **S209**). For example, since the check feature points and the second variable feature points are under the same coordinate reference, by comparing whether the coordinates of the check feature points are matched with the coordinates of the second variable feature points, it is determined whether the second original image **30** is a spoofing image or not. Specifically, when the coordinates of the check feature points are matched with the coordinates of the second variable feature points, it represents that the expression of the face in the second original image **30** has not changed and the face has not rotated, then the processor **13** determines that the second original image **30** is a spoofing image. In other words, the second original image **30** may be obtained by impersonating the user through a photo, thus there is no expression change and no face rotation. When the coordinates of the check feature points do not match with the coordinates of the second variable feature points, it represents that the expression of the face in the second original image **30** has changed or the face has rotated, then the processor **13** determines that the second original image **30** is a true man image.

(27) In some embodiments of step **S209**, the check feature points and the second variable feature points may be multiple. The processor **13** determines whether the second original image **30** is a spoofing image according to multiple check feature points and the second variable feature points corresponding to the same facial feature. The multiple check feature points and the second variable feature points corresponding to the same facial feature may be part of the check feature points and second variable feature points, or all of the check feature points and second variable feature points.

(28) In some embodiments of step **S209**, the processor **13** determines that the second original image **30** is a spoofing image when a difference degree between at least one check feature point and at least one second variable feature point is smaller than a difference threshold. Specifically, when the difference degree is smaller than the difference threshold, it represents that the expression of the face in the second original image **30** has not changed and the face has not rotated. In other

words, the second original image **30** may be obtained by impersonating the user through a photo (i.e., a spoofing image). The difference degree may be the Euclidean distance between the check feature points and the second variable feature points under the same coordinate reference. In some embodiments, the difference degree may be the Euclidean distance between the check feature points and the second variable feature points corresponding to the same facial feature under the same coordinate reference. The difference threshold may be stored in the non-transitory storage medium **11** in advance.

(29) In some embodiments, step **S205-S209** may be implemented by a correlation operation of a similarity constraint.

(30) In some embodiments, the first transform function can be calculated through simple operations (such as addition, subtraction, multiplication, and division), thus it can reduce the computing resources and memory resources needed for determining whether the second original image **30** is a spoofing image.

(31) Refer to FIG. 5, illustrating a schematic flow diagram of a liveness detection method according to some embodiments of the present application. Since steps **S501** and **S503** are the same as steps **S201** and **S203**, thus will not be repeated. In some embodiments, after obtaining the first feature points and the second feature points, the processor **13** obtains a transform function (hereinafter referred to as a fourth transform function) according to the first feature points and the second feature points (step **S505**). The fourth transform function can be implemented by a transformation matrix. For example, the processor **13** performs linear transformation on the first feature points and the second feature points to generate the fourth transform function. Specifically, the processor **13** can calculate the fourth transform function according to equation 11. Wherein, $I_{sub.2}$ is the coordinates of the second feature points, and $I_{sub.1}$ is the coordinates of the first feature points. T is the fourth transform function (e.g., an $m \times n$ transformation matrix, where m and n are 2 under two-dimensional coordinates, and m and n are 3 under homogeneous coordinates). In some embodiments of step **S505**, the processor **13** obtains the fourth transform function according to multiple first feature points and second feature points corresponding to the same facial feature. The multiple first feature points and second feature points corresponding to the same facial feature may be part of the first feature points and second feature points or all of the first feature points and second feature points.

$$I_{sub.2} = TI_{sub.1} \quad \text{Equation (11)}$$

(32) Next, the processor **13** determines whether the fourth transform function meet a predetermined constraint (step **S507**). The predetermined constraint comprises an epipolar constraint and an affine constraint. When the fourth transform function meets any one of the constraints of the predetermined constraint, the processor **13** determines that the second original image **30** is a spoofing image (step **S509**). When the fourth transform function does not meet the predetermined constraint (specifically, when the fourth transform function does not meet all the constraints of the predetermined constraint), the processor **13** determines that the second original image **30** is a true man image (step **S511**). For example, when the fourth transform function meets the epipolar constraint (as shown in equation 12), the second original image **30** is the epipolar geometry of the first original image **20**, and compared with the first original image **20**, the expression of the face in the second image **30** has not changed. Specifically, the second original image **30** may be obtained by impersonating the user through a mask, thus there is no expression change, and the processor **13** determines that the second original image **30** is a spoofing image. For another example, when the fourth transform function meets the affine constraint (as shown in equation 13 or equation 14), the second original image **30** is the affine geometry of the first original image **20**, and compared with the first original image **20**, the expression of the face in the second image **30** has not changed and the face has not rotated. Specifically, the second original image **30** may be obtained by impersonating the user through a photo, thus there is no expression change and no face rotation, and the processor **13** determines that the second original image **30** is a spoofing image.

$$I_{\text{sub.2.sup.T}} \times t \times R I_{\text{sub.1}} = 0 \quad \text{Equation (12)}$$

(33) Wherein, $I_{\text{sub.2}}$ is the coordinates of the second feature points, $I_{\text{sub.1}}$ is the coordinates of the first feature points, $I_{\text{sub.2.sup.T}}$ is a transpose matrix of $I_{\text{sub.2}}$, t is a translation vector, and R is a rotation matrix.

$$(34) \begin{bmatrix} cx \cdot \text{Math.}(1 - \cos \theta) - cy \cdot \text{Math.} \sin \theta \\ -cx \cdot \text{Math.} \sin \theta + (1 - \cos \theta) \cdot \text{Math.} cy \end{bmatrix} \quad \text{Equation(13)}$$

$$\begin{bmatrix} cx \cdot \text{Math.}(1 - \cos \theta) - cy \cdot \text{Math.} \sin \theta \\ -cx \cdot \text{Math.} \sin \theta + (1 - \cos \theta) \cdot \text{Math.} cy \end{bmatrix} \quad \text{Equation(14)} \quad = \text{scl} * \cos \theta \quad \text{Equation(15)}$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \text{scl} * \sin \theta \quad \text{Equation(16)}$$

(35) Wherein, cx and cy are the horizontal and vertical coordinates of the rotation center, respectively, scl is a scaling scale, and θ is the rotation radian.

(36) When the fourth transform function does not meet the predetermined constraint, it represents that the second original image **30** is not a kind of geometric deformation of the first original image **20**. In other words, the expression of the face in the second original image **30** may have changed. That is, the second original image **30** is photographed by a real person. Therefore, under this condition, the processor **13** determines that the second original image **30** is a true man image.

(37) In some embodiments, the fourth transform function can be calculated through simple operations (such as addition, subtraction, multiplication, and division), thus it can reduce the computing resources and memory resources needed for determining whether the second original image **30** is a spoofing image.

(38) In some embodiments, the first transform function is different from the fourth transform function. In other words, different determination conditions can be respectively formed based on the first transform function and the fourth transform function (e.g., steps **S209** and **S507**). In this way, the accuracy of detecting whether the second original image **30** is a spoofing image can be improved.

(39) Refer to FIG. **6**, illustrating a schematic flow diagram of a liveness detection method according to some embodiments of the present application. In some embodiments, before obtaining the first feature points and the second feature points (or, before obtaining the first fixed feature points and the at least one first variable feature point and before obtaining the second fixed feature points and the at least one second variable feature point), the processor **13** performs steps **S600-S606**. First, the processor **13** obtains a first authentication result according to a first deskewed image and a facial recognition process (step **S600**). Next, the processor **13** obtains a second authentication result according to a second deskewed image and the facial recognition process (step **S602**).

(40) The first deskewed image is obtained by performing a deskew process (described in detail later) on the first original image **20**, and the second deskewed image is obtained by performing the deskew process on the second original image **30**. For example, a frame before the deskew process is an original image **40** (as shown in FIG. **9A**), and the face in the original image **40** may be too small, too large, or oblique. If the facial recognition is directly performed on the original image **40**, it may cause the error of facial recognition to increase. The deskew process can appropriately enlarge, reduce, and rotate the original image **40** to generate a deskewed image **50** with the face in a proper size and correction direction (as shown in FIG. **9C**). For example, the face in the deskewed image **50** is not oblique, occupies two-thirds of the layout of the deskewed image **50**, and is located in the center of the deskewed image **50**. In this way, the accuracy of the facial recognition process can be improved.

(41) The facial recognition process can be implemented by a convolutional neural network. For

example, the designer can input multiple sample images with marked feature points and the face has been deskewed (hereinafter referred to as feature-point-deskewed sample images) to the processor **13** through the I/O interface. The processor **13** performs the machine learning training of facial recognition according to the feature-point-deskewed sample images, so as to determine a determination logic (hereinafter referred to as third determination logic). The machine learning training of facial recognition may be a known or self-developed model, and the details are omitted here. According to the third determination logic, the processor **13** can perform facial recognition on the images (such as the first deskewed image and the second deskewed image) to generate the facial description vectors associated with the images (such as a first facial description vector associated with the first deskewed image and a second facial description vector associated with the second deskewed image). Wherein, each face corresponds to one facial description vector, and different faces correspond to different facial description vectors. In other words, if the face in the first deskewed image is different from the face in the second deskewed image, the first facial description vector is different from the second facial description vector. In some embodiments, the facial description vectors can be implemented by using floating numbers. For example, the facial description vectors may be 128-bit or 256-bit floating numbers. In some embodiments, the first determination logic, the second determination logic, and the third determination logic are different from each other.

(42) The non-transitory storage medium **11** can store a plurality of registered facial description vectors, and the registered facial description vectors respectively correspond to a plurality of users who have been registered to the liveness detection system **10** (hereinafter referred to as registered users). The processor **13** determines whether the first facial description vector matches any one of the registered facial description vectors in the non-transitory storage medium **11**. If the first facial description vector matches any one of the registered facial description vectors, it represents that the face in the first deskewed image matches the face of one of the registered users, and the processor **13** uses the first facial description vector as the first authentication result. If the first facial description vector does not match any one of the registered facial description vectors, it represents that the face in the first deskewed image does not match the face of any registered users, the processor **13** generates an authentication failure instruction as the first authentication result.

(43) Similar to the first authentication result, the processor **13** determines whether the second facial description vector matches any one of the registered facial description vectors in the non-transitory storage medium **11**. If the second facial description vector matches any one of the registered facial description vectors, it represents that the face in the second deskewed image matches the face of one of the registered users, and the processor **13** uses the second facial description vector as the second authentication result. If the second facial description vector does not match any one of the registered facial description vectors, it represents that the face in the second deskewed image does not match the face of any registered users, the processor **13** generates an authentication failure instruction as the second authentication result.

(44) In some embodiments, the processor **13** can perform a Euclidean distance operation or a cosine similarity operation on the first facial description vector and the registered facial description vectors (or the second facial description vector and the registered facial description vectors) to obtain a comparison parameter. When the comparison parameter is less than a comparison threshold, it represents that the first facial description vector matches the registered facial description vector (or the second facial description vector matches the registered facial description vector). When the comparison parameter is not less than the comparison threshold, it represents that the first facial description vector does not match the registered facial description vectors (or the second facial description vector does not match the registered facial description vectors). The comparison parameter can be stored in the non-transitory storage medium **11** in advance.

(45) After obtaining the first authentication result and the second authentication result, the processor **13** can determine whether the first authentication result is the same as the second

authentication result (step **S606**). For example, the processor **13** determines whether the first facial description vector of the first authentication result is the same as the second facial description vector of the second authentication result. In other words, the processor **13** determines whether the first facial description vector and the second facial description vector match the facial description vector of the same registered user. When the first authentication result is the same as the second authentication result (e.g., the first facial description vector is the same as the second facial description vector), the processor **13** performs the steps of obtaining the first feature points and the second feature points (i.e., steps **S201-S203** or steps **S501-S503**). When the first authentication result is different from the second authentication result, the processor **13** repeats the step of obtaining the first authentication result (i.e., step **S600**). For example, the processor **13** controls the image capture device **15** to generate another video signal having another set of frames, and the processor **13** re-obtains another first deskewed image according to the another set of frames. The processor **13** performs the facial recognition process on the another first deskewed image to generate a new first authentication result and performs step **S602** and steps following it according to the another set of frames. In this way, it can be ensured that the detection of a real or fake person is performed under the condition that the second original image **30** is photographed from a registered user. That is to say, the computing resources and memory resources consumed by the processor **13** for the detection of a real or fake person can be reduced (for example, the detection of a real or fake person does not need to be performed under the condition that the second original image **30** is not photographed from a registered user).

(46) Refer to FIG. 7, illustrating a schematic flow diagram of a liveness detection method according to some embodiments of the present application. In some embodiments, when the first authentication result is different from the second authentication result, the processor **13** repeats the step of obtaining the second authentication result (i.e., step **S602**). For example, the processor **13** reselects another second original image **30** from the frames following the first original image **20**, performs the deskew process on the another second original image **30** to generate another second deskewed image, re-performs the facial recognition process on the another second deskewed image to generate a new second authentication result, and performs step **S604**.

(47) In step **S604**, the processor **13** determines whether a repetition number of the step of obtaining the second authentication result (i.e., step **S602**) reaches a predetermined number (step **S604**). When the repetition number of the step of obtaining the second authentication result has not reached the predetermined number, the processor **13** performs step **S606**. When the repetition number of the step of obtaining the second authentication result reaches the predetermined number, it represents that it has performed multiple times of determination of step **S606** according to multiple second authentication results of the second deskewed images, and the processor **13** repeats the step of obtaining the first authentication result (i.e., step **S600**). In other words, under the condition that step **S606** has been performed multiple times, the first authentication result is still different from the second authentication result. The predetermined number can be stored in the non-transitory storage medium **11** in advance.

(48) For example, under some conditions, a certain user leaves the scene immediately after using the liveness detection system **10**, and another user uses the liveness detection system **10** after the previous user leaves the scene. That is to say, in this case, under the same set of frames that were captured, the face of the first deskewed image generated based on the first original image **20** is different from the face of the second deskewed image generated based on the second original image **30**, such that the first facial description vector of the first deskewed image served as the first authentication result is different from the second facial description vector of the second deskewed image served as the second authentication result. Therefore, the determination number of step **S606** can be limited through step **S604**, thereby saving the computing resources of the processor **13**, and ensuring that the user using the liveness detection system **10** will not change within a certain time.

(49) In some embodiments, when the first authentication result has an authentication failure

instruction, the processor **13** repeats the step of obtaining the first authentication result (i.e., step **S600**). Specifically, when the first authentication result has an authentication failure instruction, it represents that the face in the first deskewed image does not correspond to (photograph from) the face of any one of the registered users. Therefore, the processor **13** repeats step **S600** until the first authentication result has a first facial description vector (or the first authentication result does not have an authentication failure instruction). Wherein, the repetition manner of step **S600** has been described above, and will not be repeated here. Similarly, when the second authentication result has the authentication failure instruction, the processor **13** repeats the step of obtaining the second authentication result (i.e., step **S602**). Specifically, when the second authentication result has an authentication failure instruction, it represents that the face in the second deskewed image does not correspond to (photograph from) the face of any one of the registered users. Therefore, the processor **13** repeats step **S602** until the second authentication result has a second facial description vector (or the second authentication result does not have an authentication failure instruction). Wherein, the repetition manner of step **S602** has been described above, and will not be repeated here.

(50) Refer to FIG. 8, illustrating a schematic flow diagram of a deskew process according to some embodiments of the present application. In some embodiments, when performing the deskew process, the processor **13** obtains a transform function (hereinafter referred to as a second transform function) according to the first fixed feature points and the two preset feature points (step **S800**). Specifically, the processor **13** obtains the second transform function according to the coordinates of the first fixed feature points and the preset feature points. That is to say, the coordinates of the first fixed feature points can be converted to the coordinates of the preset feature points through the second transform function. The two preset feature points can be used as alignment points for generating the first deskewed image. The two preset feature points can be designed by the designer and stored in the non-transitory storage medium **11**. In some embodiments, two first fixed feature points and two preset feature points correspond to the same facial feature, for example, the two first fixed feature points and two preset feature points both correspond to two eyeballs of the face. Then, the processor **13** performs face alignment on the first original image **20** according to the second transform function and each pixel in the first original image **20** to convert the first original image **20** into the first deskewed image (step **S802**). For example, the processor **13** converts the coordinates of each pixel of the first original image **20** into new coordinates through the second transform function and arranges each pixel of the first original image **20** according to each new coordinate to form the first deskewed image. In some embodiments, the second transform function may be a similarity transformation matrix.

(51) Similar to steps **S800-S802**, when performing the deskew process, the processor **13** further obtains a transform function (hereinafter referred to as a third transform function) according to the second fixed feature points and the preset feature points (step **S804**). Then, the processor **13** performs face alignment on the second original image **30** according to the third transform function and each pixel in the second original image **30** to convert the second original image **30** into the second deskewed image (step **S806**). In some embodiments, the order of steps **S800-S802** and steps **S804-806** may be interchanged.

(52) In some embodiments of step **S804**, the processor **13** obtains the third transform function according to the coordinates of the second fixed feature points and the preset feature points. That is to say, the coordinates of the second fixed feature points can be converted to the coordinates of the preset feature points through the third transform function. In addition to being used as alignment points for generating the first deskewed image, the two preset feature points can further be used as alignment points for generating the second deskewed image. In some embodiments of step **S804**, two second fixed feature points and two preset feature points correspond to the same facial feature, for example, the two second fixed feature points and two preset feature points both correspond to two eyeballs of the face. In some embodiments of step **S806**, the processor **13** converts the

coordinates of each pixel of the second original image **30** into new coordinates through the third transform function and arranges each pixel of the second original image **30** according to each new coordinate to form the second deskewed image. In some embodiments, the third transform function may be a similarity transformation matrix.

(53) Refer to FIGS. **9A-9C**, illustrating a schematic diagram of an original image **40** and a deskewed image **50** according to some embodiments of the present application. It can be seen from FIG. **9A** that two fixed feature points O.sub.DD and A.sub.DD (such as the two first fixed feature points or the two second fixed feature points) can be obtained through performing a feature point process on the original image **40** (such as the first original image **20** or the second original image **30**). Next, according to the two fixed feature points O.sub.DD and A.sub.DD and two preset feature points O.sub.SS and A.sub.SS (as shown in FIG. **9B**), a transform function between the two fixed feature points O.sub.DD and A.sub.DD and two preset feature points O.sub.SS and A.sub.SS (such as the second transform function or the third transform function) can be calculated. Then, the deskewed image **50** (such as the first deskewed image or the second deskewed image) (as shown in FIG. **9C**) can be generated by converting each pixel in the original image **40** through the transform function.

(54) In some embodiments, the second transform function and the third transform function are substantially the same as the first transform function. For example, in equation 1 to equation 8, O.sub.S and A.sub.S may be the preset feature points, and O.sub.D and A.sub.D may be the two first fixed feature points or the two second fixed feature points. In this way, the computing resources and storage resources of the liveness detection system **10** can be saved. For example, the first transform function, the second transform function, and the third transform function can be implemented only based on a kind of transformation matrix. In some embodiments, since the second transform function and the third transform function are substantially the same as the first transform function and the first transform function is different from the fourth transform function, the second transform function and the third transform function are different from the fourth transform function.

(55) To sum up, according to some embodiments, the first transform function and the fourth transform function can be obtained through a simple operation, and whether a person image to be recognized is a spoofing image can be determined according to the first transform function and the fourth transform function, thereby determining whether the behavior of using photos or videos for spoofing occurs. In other words, while realizing the determination of the spoofing behavior, the computing resources and memory resources required for performing the determination of the spoofing behavior can also be reduced.

Claims

1. A liveness detection method, implemented in a facial recognition technology, comprising: obtaining, according to a first image and a feature point process, a plurality of first feature points; selecting two first fixed feature points and at least one first variable feature point from the plurality of first feature points; obtaining, according to a second image and the feature point process, a plurality of second feature points; selecting two second fixed feature points and at least one second variable feature point from the plurality of second feature points; wherein the variable feature points are more likely to change with expressions than the fixed feature points; obtaining, according to each pixel of the first fixed feature points and each pixel of the second fixed feature points, a first transform function; obtaining, according to the first transform function and each pixel of the at least one first variable feature point, at least one check feature point; and determining, according to the at least one check feature point and the at least one second variable feature point, whether the second image is a spoofing image.

2. The liveness detection method according to claim 1, wherein when a difference degree between

the at least one check feature point and the at least one second variable feature point is less than a difference threshold, determining that the second image is the spoofing image.

3. The liveness detection method according to claim 1, wherein the first transform function is

$$O_D A_D = [\begin{smallmatrix} rs0 & -rs1 \\ rs1 & rs0 \end{smallmatrix}] (\overset{.fwdarw.}{O_S A_S} + [\begin{smallmatrix} t_x \\ t_y \end{smallmatrix}])$$
 and is substantially the same as

$$[\begin{smallmatrix} dc_x \\ dc_y \end{smallmatrix}] = [\begin{smallmatrix} rs0 & -rs1 \\ rs1 & rs0 \end{smallmatrix}] [\begin{smallmatrix} sc_x \\ sc_y \end{smallmatrix}] + [\begin{smallmatrix} t_x \\ t_y \end{smallmatrix}],$$
 and wherein O.sub.s and A.sub.s are the first fixed feature

points, O.sub.D and A.sub.D are the second fixed feature points, t.sub.x is a horizontal translation amount of O.sub.s and A.sub.s, t.sub.y is a vertical translation amount of O.sub.s and A.sub.s,

$rs0 = \frac{ds_x * dd_x + ds_y * dd_y}{ds_x^2 + ds_y^2}$, $rs1 = \frac{ds_x * dd_y - ds_y * dd_x}{ds_x^2 + ds_y^2}$, ds.sub.x is a horizontal component of the vectors of

O.sub.s and A.sub.s, ds.sub.y is a vertical component of the vectors of O.sub.s and A.sub.s,

dd.sub.x is a horizontal component of the vectors of the vectors of O.sub.D and A.sub.D, dd.sub.y

is a vertical component of the vectors of O.sub.D and A.sub.D, dc.sub.x is a horizontal coordinate

of the at least one check feature point, dc.sub.y is a vertical coordinate of the at least one check

feature point, sc.sub.x is a horizontal coordinate of the at least one first variable feature point, and

sc.sub.y is a vertical coordinate of the at least one first variable feature point.

4. The liveness detection method according to claim 3, wherein the first transform function is

$$\begin{matrix} dc_x & rs0 & -rs1 & t_x & sc_x \\ \text{substantially the same as } [\begin{smallmatrix} dc_y \end{smallmatrix}] = [\begin{smallmatrix} rs1 & rs0 & t_y \end{smallmatrix}] [\begin{smallmatrix} sc_y \end{smallmatrix}] . \\ 1 & 0 & 0 & 1 & 1 \end{matrix}$$

5. The liveness detection method according to claim 1, further comprising: before obtaining the first fixed feature points and the at least one first variable feature point and before obtaining the second fixed feature points and the at least one second variable feature point, executing: obtaining, according to a first deskewed image and a facial recognition process, a first authentication result, wherein the first deskewed image is obtained by performing a deskew process on the first image; obtaining, according to a second deskewed image and the facial recognition process, a second authentication result, wherein the second deskewed image is obtained by performing the deskew process on the second image; and repeating the step of obtaining the first authentication result when the first authentication result is different from the second authentication result.

6. The liveness detection method according to claim 5, wherein when the first authentication result is the same as the second authentication result, performing the step of obtaining the first fixed feature points and the at least one first variable feature point, and the step of obtaining the second fixed feature points and the at least one second variable feature point.

7. The liveness detection method according to claim 5, wherein when the first authentication result has an authentication failure instruction, repeating the step of obtaining the first authentication result, and when the second authentication result has the authentication failure instruction, repeating the step of obtaining the second authentication result.

8. The liveness detection method according to claim 5, wherein the deskew process comprises: obtaining, according to the first fixed feature points and two preset feature points, a second transform function; performing, according to the second transform function and each pixel in the first image, a face alignment on the first image to convert the first image into the first deskewed image; obtaining, according to the second fixed feature points and the preset feature points, a third transform function; and performing, according to the third transform function and each pixel in the second image, the face alignment on the second image to convert the second image into the second deskewed image, wherein the second transform function and the third transform function are substantially the same as the first transform function.

9. The liveness detection method according to claim 1, further comprising: obtaining, according to the first feature points and the second feature points, a fourth transform function; and determining

the second image is a liveness image when the fourth transform function does not meet a predetermined constraint, wherein the predetermined constraint comprises an epipolar constraint and an affine constraint.

10. The liveness detection method according to claim 9, further comprising: determining the second image is the spoofing image when the fourth transform function meets any one of the constraints in the predetermined constraint.

11. The liveness detection method according to claim 9, further comprising: before obtaining the plurality of first feature points and the plurality of second feature points, performing: obtaining, according to a first deskewed image and a facial recognition process, a first authentication result, wherein the first deskewed image is obtained by performing a deskew process on the first image; obtaining, according to a second deskewed image and the facial recognition process, a second authentication result, wherein the second deskewed image is obtained by performing the deskew process on the second image; performing the steps of obtaining the plurality of first feature points and the plurality of second feature points when the first authentication result is the same as the second authentication result; and repeating the step of obtaining the first authentication result when the first authentication result is different from the second authentication result.

12. The liveness detection method according to claim 11, wherein the deskew process comprises: obtaining, according to two first fixed feature points in the plurality of first feature points and two preset feature points, a second transform function; performing, according to the second transform function and each pixel in the first image, a face alignment on the first image to convert the first image into the first deskewed image; obtaining, according to two second fixed feature points in the plurality of second feature points and the preset feature points, a third transform function; and performing, according to the third transform function and each pixel in the second image, the face alignment on the second image to convert the second image into the second deskewed image, wherein the second transform function and the third transform function are different from the fourth transform function.

13. A liveness detection system, implemented in a facial recognition technology, comprising: a non-transitory storage medium, configured to store a program; and a processor, electrically connected to the non-transitory storage medium, configured to load the program to perform the following steps: obtaining, according to a first image and a feature point process, a plurality of first feature points; selecting two first fixed feature points and at least one first variable feature point from the plurality of first feature points; obtaining, according to a second image and the feature point process, a plurality of second feature points; selecting two second fixed feature points and at least one second variable feature point from the plurality of second feature points; wherein the variable feature points are more likely to change with expressions than the fixed feature points; obtaining, according to each pixel of the first fixed feature points and each pixel of the second fixed feature points, a first transform function; obtaining, according to the first transform function and each pixel of the at least one first variable feature point, at least one check feature point; and determining, according to the at least one check feature point and the at least one second variable feature point, whether the second image is a spoofing image.

14. The liveness detection system according to claim 13, wherein when a difference degree between the at least one check feature point and the at least one second variable feature point is less than a difference threshold, the processor determines that the second image is the spoofing image.

15. The liveness detection system according to claim 13, wherein the first transform function is

$$O_D A_D = \begin{bmatrix} rs0 & -rs1 \\ rs1 & rs0 \end{bmatrix} (O_S A_S + \begin{bmatrix} t_x \\ t_y \end{bmatrix})$$
 and is substantially the same as

$$\begin{bmatrix} dc_x \\ dc_y \end{bmatrix} = \begin{bmatrix} rs0 & -rs1 \\ rs1 & rs0 \end{bmatrix} \begin{bmatrix} sc_x \\ sc_y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix},$$
 and wherein O.sub.s and A.sub.s are the first fixed feature

points, O.sub.D and A.sub.D are the second fixed feature points, t.sub.x is a horizontal translation

amount of O.sub.s and A.sub.s, t.sub.y is a vertical translation amount of O.sub.s and A.sub.s,
 $rs0 = \frac{ds_x * dd_x + ds_y * dd_y}{ds_x^2 + ds_y^2}$, $rs1 = \frac{ds_x * dd_y - ds_y * dd_x}{ds_x^2 + ds_y^2}$, ds.sub.x is a horizontal component of the vectors of
O.sub.s and A.sub.s, ds.sub.y is a vertical component of the vectors of O.sub.s and A.sub.s,
dd.sub.x is a horizontal component of the vectors of the vectors of O.sub.D and A.sub.D, dd.sub.y
is a vertical component of the vectors of O.sub.D and A.sub.D, dc.sub.x is a horizontal coordinate
of the at least one check feature point, dc.sub.y is a vertical coordinate of the at least one check
feature point, sc.sub.x is a horizontal coordinate of the at least one first variable feature point, and
sc.sub.y is a vertical coordinate of the at least one first variable feature point.

16. The liveness detection system according to claim 13, wherein the first transform function is

$$\begin{matrix} & dc_x & & rs0 & -rs1 & t_x & & sc_x \\ \text{substantially the same as } [& dc_y &] = [& rs1 & rs0 & t_y &][& sc_y &]. \\ & 1 & & 0 & 0 & 1 & & 1 \end{matrix}$$

17. The liveness detection system according to claim 13, wherein before obtaining the first fixed
feature points and the at least one first variable feature point and before obtaining the second fixed
feature points and the at least one second variable feature point, the processor performs the
following steps: obtaining, according to a first deskewed image and a facial recognition process, a
first authentication result, wherein the first deskewed image is obtained by performing a skew
process on the first image; obtaining, according to a second deskewed image and the facial
recognition process, a second authentication result, wherein the second deskewed image is obtained
by performing the skew process on the second image; and repeating the step of obtaining the first
authentication result when the first authentication result is different from the second authentication
result.

18. The liveness detection system according to claim 17, wherein when the first authentication
result is the same as the second authentication result, the processor performs the step of obtaining
the first fixed feature points and the at least one first variable feature point, and the step of
obtaining the second fixed feature points and the at least one second variable feature point.

19. The liveness detection system according to claim 17, wherein when the first authentication
result has an authentication failure instruction, the processor repeats the step of obtaining the first
authentication result, and when the second authentication result has the authentication failure
instruction, the processor repeats the step of obtaining the second authentication result.

20. The liveness detection system according to claim 17, wherein the skew process comprises:
obtaining, according to the first fixed feature points and two preset feature points, a second
transform function; performing, according to the second transform function and each pixel in the
first image, a face alignment on the first image to convert the first image into the first skewed
image; obtaining, according to the second fixed feature points and the preset feature points, a third
transform function; and performing, according to the third transform function and each pixel in the
second image, the face alignment on the second image to convert the second image into the second
skewed image, wherein the second transform function and the third transform function are
substantially the same as the first transform function.
